

Calibrated Query Augmentation and Dual Preference Alignment for Retrieval-Augmented Generation

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Abstract—Even though Retrieval-Augmented Generation (RAG) models have shown promise in integrating outside knowledge into Large Language Models, there is still a fundamental problem in this field that needs to be solved. Retrieval models that were created with traditional IR metrics at their core have a tendency to perform poorly on LLM tasks. However, there is a fundamental performance bottleneck because documents that are reranked using conventional relevance metrics frequently lack the supporting context needed to produce the right answer. In this paper, we propose a preference-based learning paradigm, CQADPA-RAG, to rectify this misalignment. We perform a multi-dimensional query augmentation with the following structure: We start by rephrasing the queries to make them more clear on one end, adding causal dependencies on the other, taking into account several constraints in the middle, and using scoring to verify the citation requirements. We employ confidence-weighted labeling across multiple models to guarantee high-quality annotations. Our solution's two-stage training mechanism is its key component. Multi-granular preference learning is used in the first stage to take advantage of classification while introducing contrastive losses within and between queries for retrieval at various scales. In the second stage, we incorporate bidirectional LLM preference alignment, a dual ranking mechanism that not only identifies beneficial documents but also blocks harmful ones. Our analysis of five QA benchmarks demonstrates that our model consistently outperforms the current models in terms of support inference and answer accuracy. We demonstrate through systematic ablation studies that every design decision is crucial to attaining such a high performance, particularly for complex reasoning.

Index Terms—Retrieval-Augmented Generation, Preference Alignment, Large Language Models

I. INTRODUCTION

The rise of Large Language Models (LLMs) has completely changed the way computers process natural language [1]–[3]. Machines can understand and write text that sounds like a person with an amazing level of sophistication. But a big problem with these models is that their knowledge stops growing when they finish training. All that they know is contained in their training data, so there's no way for them to gain access to information or trends that come about after the fact. This inability has several detrimental effects. Temporal knowledge gaps becomes wider through time, the models sometimes produce realistic-sounding but completely fabricated facts when subjected to knowledge-intensive queries [4] [5]. Retrieval-Augmented Generation (RAG) [6], [7] has been proposed in an attempt to address this Challenging setting. Instead of making LLMs depend entirely on the internal, static knowledges they have, RAG systems fetch dynamic and potentially relevant information from external sources or knowledge bases such as Wikipedia to generate a response dynamically [8]. Taking the generation based on knowable external facts effectively resolves various issues with the model.

However, RAG systems themselves suffer from fundamental inconsistencies. Current retrieval models are predominantly optimized for traditional information retrieval metrics such as relevance and semantic similarity [9], yet these metrics do not necessarily reflect the actual needs of LLMs when generating answers. A document may contain information relevant to the topic and score high in semantic similarity, yet fail to support correct reasoning. Furthermore, retrieved documents may contradict the LLM's internal knowledge, potentially

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undermining its otherwise functional reasoning capabilities [10], [11]. Recent research has elucidated the influence of documents on LLMs from various viewpoints [9], [12], [13], indicating that documents with accurate answers can still mislead LLMs into producing erroneous responses if they contradict the model’s reasoning preferences. This discovery highlights the necessity of resolving the consistency disparity between retrieval models and LLMs.

Numerous ideas have been put forth to improve RAG systems, but the issues remain unresolved. Reranking techniques rearrange retrieved documents using LLM-based rankers or dense retrievers. These techniques primarily focus on improving rankers’ relevance scores rather than determining what LLMs actually desire. To improve the alignment between retrieval and generation, preference alignment techniques [14]–[17] make use of ideas from Reinforcement Learning from Human Feedback (RLHF) [18], [19]. However, these approaches typically only identify documents that people find useful and learn which ones they like. They do not model preferences for negative documents, nor do they specify which documents should be filtered out. In order to cover more ground and improve RAG’s overall performance, data augmentation strategies [20], [21] generate training queries. However, they typically produce data that isn’t very complex or diverse, which doesn’t truly reflect what people need. When it comes to preference annotation during data labeling, these methods lack robust quality control, which could harm model training by adding noise or creating inconsistent labels.

Recent research on RAG has branched out into new areas, such as agentic search [22], adaptive retrieval strategies [23], and graph-based RAG methods [24]. Agentic search refines queries by going through the same steps over and over again. Adaptive retrieval changes how it retrieves information based on the situation. Graph-based methods use structured knowledge representations to make it easier to find information. Even though these methods make retrieval better, they all focus on making the retrieval process better without addressing the mismatch between retrieval targets and the generation requirements of LLM. For example, graph-based methods still use standard relevance metrics instead of LLM-specific preferences. As a result, they have trouble figuring out the exact generation needs that LLMs really need and don’t deal with the important asymmetry problems we’ve seen. LLMs are very good at finding useful documents but not very good at finding and filtering out useless or wrong documents [9]. This causes the model to not be able to filter out wrong answers, which leads to misleading results and lowers the model’s overall performance and reliability.

We propose CQADPA-RAG (Calibrated Query Augmentation and Dual Preference Alignment for RAG) to align the different knowledge preferences of LLMs in RAG systems using two new ideas. (1) Multi-Dimensional Query Augmentation with Confidence-Calibrated labeling. We create four query augmentation strategies—basic reformulation, causal reasoning enhancement, multi-constraint specification, and citation verification—each based on what real users want.

We present confidence-calibrated labeling, which uses multi-model ensemble and automatic quality filtering to create high-quality preference data. This solves the problem of not having enough data in previous methods. (2) Multi-Granular and Bidirectional LLM Preference Alignment. We employ a two-stage training framework to synchronise RAG systems with LLM preferences. The multi-granular preference learning stage records preferences at three levels: instance-level classification learns preferences for single documents; pair-level contrastive learning records relative preferences within each query to tell useful documents from useless ones; and set-level contrastive learning finds generalisable preferences across different queries. In the bidirectional LLM preference alignment stage, we model and learn both positive and negative preferences of LLMs. Forward alignment raises the rank of useful documents, while backward alignment actively lowers the rank of misleading ones. This bidirectional optimisation fully fixes the problem with current systems where positive and negative rankings are not equal.

We perform comprehensive experiments on five public QA datasets to validate our method: CLAPNQ [25], HotpotQA [26], MS_MARCO [27], Natural Questions [28], and TriviaQA [29]. Results show that both the accuracy of the answers and the rate of support for the answers have improved a lot compared to the best baselines. Extensive ablation studies indicate that both query augmentation and dual preference alignment significantly enhance overall performance.

Our contributions can be summarized as follows:

- We introduce a multi-dimensional query augmentation strategy with confidence-calibrated labeling that constructs high-quality training queries from four complementary perspectives. Using multi-model ensemble for confidence labeling effectively alleviates data scarcity.
- We develop a two-stage preference alignment framework that captures retrieval preferences at multiple granularities (instance, pair, and set levels) and addresses the positive-negative ranking asymmetry in existing RAG systems by modeling and learning bidirectional LLM preferences.
- We achieve significant improvements over strong baselines through extensive experiments on five QA datasets in both answer accuracy and answer support rate. Ablation experiments also validate the importance of each component to the framework.

II. RELATED WORK

A. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) has emerged as a significant method for mitigating the limitations of LLMs in knowledge-intensive tasks. Lewis et al. [6] were the first to talk about RAG as a way to improve generation with retrieved passages. REALM [7] put forward a pre-training method that trains both the retriever and the generator at the same time. Recent research has investigated various avenues. Agentic search frameworks [22] iteratively enhance retrieval strategies. Metacognitive approaches [23] facilitate models in executing

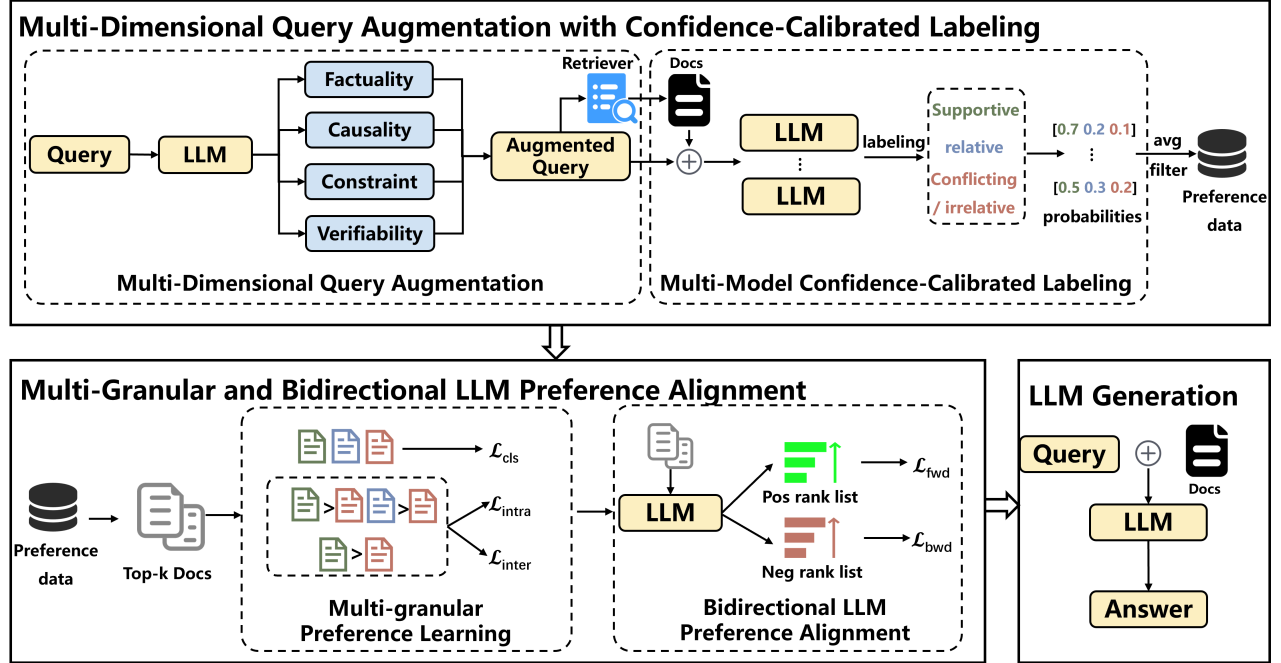


Fig. 1. Process of Calibrated Query Augmentation and Dual Preference Alignment for Retrieval-Augmented Generation.

self-reflective retrieval decisions. Graph-based methods [24] employ structured knowledge representations. However, there is a fundamental issue with all of these methods: retrieval models that are optimised for traditional IR metrics do not capture what LLMs actually need for accurate generation [9]. This discrepancy between retrieval goals and generation needs is the focus of our research.

B. Query Augmentation and Data Construction

Query augmentation is important for improving retrieval quality. By expanding or reformulating user queries, systems can better understand what users actually need. Early work focused on synthetic query generation using LLMs [20]. Zero-shot approaches [21] showed dense retrieval could work without relevance labels. Recent work in instruction tuning [30]–[32] find that organizing training data by task types improves model capabilities. Instruction synthesis methods [33]–[36] showed that diversity in constructed data matters.

Research in multi-view learning has investigated the utilisation of various complementary perspectives. Different perspectives offer supplementary information that enhances performance. This operates via view fusion [37]–[41], managing incomplete views [42], or simulating view-category interactions [43]. These methods demonstrate that the incorporation of diverse perspectives is superior to reliance on singular viewpoints. We get ideas from both “task type organisation” in instruction tuning and “multi-perspective integration” in multi-view learning. But we use this for how we ask questions and how we interact. We don’t just rephrase questions; we also create different human query patterns from different angles.

There are two main problems with the current ways when improving RAG. They make queries from only one point of view, which doesn’t show how people really ask for information. And without proper calibration, the quality of augmented data can be very different [44]. This adds noise to the training signals and makes performance worse. Multi-dimensional query augmentation solves both problems. We build queries from four different points of view, each showing a different way that people interact with queries. We also introduce confidence-calibrated labeling to make sure the quality of labeling.

C. Preference Alignment for RAG

It is important to ensure that the retrieval system can meet the needs of LLM. REPLUG [14] was the first to use LLM feedback to improve retrievers. Subsequent research investigated domain-specific alignment [15], ranking-based feedback [16], and domain adaptation [17]. These techniques employ reinforcement learning [18] or direct preference optimisation [19] methodologies. But they have three big problems. They only work at one level of granularity, so they miss preferences at the instance, pair, and set levels. They focus on what LLMs like, or positive preferences, and ignore what LLMs should avoid, or negative preferences. And they don’t talk about how LLMs deal with positive and negative ranking in an uneven way [9], which is based on a problem that harmful documents hurt performance more than helpful documents help it. Recent studies have attempted to eliminate misleading context [12] or reconcile knowledge discrepancies [13]. These efforts were somewhat successful. However, there was still no complete

framework for bidirectional alignment. Our two-stage preference alignment method fills this gap. We collect preferences at different levels of detail and use dual listwise ranking objectives to model both positive and negative preferences.

III. METHODOLOGY

The overall framework of CQADPA-RAG is illustrated in Figure 1, which consists of three main components: (1) Multi-Dimensional Query Augmentation with Confidence-Calibrated Annotation, which is used to construct high-quality preference data; (2) Multi-Granular and Bidirectional LLM Preference Alignment, which trains a preference-aligned reranker in two stages; and (3) LLM Generation, which uses the reranked documents to generate the final answer. We detail each component in the following subsections.

A. Task Definition

In a standard RAG system, the retrieval process consists of three stages. Given a user query q and document corpus $\mathcal{D}$, a retriever $\mathcal{R}$ first identifies the top- N candidate documents $D_q = \{d_1, d_2, \dots, d_N\}$ based on relevance scores. A reranker $\mathcal{E}_r$ then reorders these candidates to select the top- k documents, where $k \ll N$:

$$D_{\text{rerank}} = \mathcal{E}_r(q, D_q) = \text{top-}k\{s_\theta(q, d_i) \mid d_i \in D_q\} \quad (1)$$

where $s_\theta(q, d_i)$ represents the reranking score for document d_i given query q . Finally, an LLM generator $\mathcal{G}$ produces the answer based on the reranked documents:

$$y = \mathcal{G}(q, D_{\text{rerank}}) = \arg \max_y P_\theta(y \mid q, D_{\text{rerank}}) \quad (2)$$

Our objective is to align the reranker $\mathcal{E}_r$ with LLM preferences, ensuring that D_{rerank} contains documents that genuinely support accurate answer generation rather than merely exhibiting semantic relevance to the query.

B. Multi-Dimensional Query Augmentation with Confidence-Calibrated Labeling

We create a multi-dimensional query augmentation framework to make high-quality training data that meets a wide range of user information needs. We don't just rephrase simple queries; we build queries from four different points of view, each showing a different way that people interact with queries:

Basic Reformulation. This dimension mimics the various ways people naturally say the same thing by rewriting the original question while maintaining its main idea. While maintaining the same meaning, this addition captures the diversity in the ways that queries are constructed.

Enhancement of causal reasoning. We modify queries to request causal explanations or chains of reasoning. For example, it is more likely that documents supporting step-by-step reasoning will be found if a factual question is changed to one like "Why did X happen?" or "What caused Y?"

Multi-Constraint Specification. We can better meet people's actual information needs which are frequently when they need it in a particular way by making queries more complex.

Citation Verification Requirements. We reword our questions to explicitly request evidence-based responses or source verification.

Given an original training set $\mathcal{D}_{\text{train}} = \{(q_i, y_i)\}_{i=1}^{N_{\text{train}}}$, we utilize GPT-4 to generate augmented queries $\{q_i^{(j)}\}_{j=1}^4$ for each dimension. We also generate corresponding augmented answers $\{y_i^{(j)}\}_{j=1}^4$ to maintain consistency between queries and expected responses. This process produces an augmented dataset:

$$\mathcal{D}_{\text{aug}} = \bigcup_{i=1}^{N_{\text{train}}} \bigcup_{j=1}^4 \{(q_i^{(j)}, y_i^{(j)})\} \quad (3)$$

To ensure high-quality preference annotations, we introduce a multi-model ensemble labeling mechanism with confidence calibration. For each query-document pair (q, d) , we leverage multiple LLMs to assess whether the document supports correct answer generation.

Multi-Model Ensemble: We employ four independent labeling attempts using two state-of-the-art models (GPT-4.1 and DeepSeek-V3.1) with two different prompt templates. Each produces label $l \in \{0, 1, 2\}$ representing:

- 0: Supportive (aligned knowledge)
- 1: Neutral/Irrelevant
- 2: Conflicting (unaligned knowledge)

Confidence Score Extraction: We extract the probability distribution from first token logits:

$$p_m(l \mid q, d) = \frac{\exp(\text{logit}_m(l))}{\sum_{l'=0}^2 \exp(\text{logit}_m(l'))} \quad (4)$$

where $m \in \{1, 2, 3, 4\}$ denotes the labeling attempt.

Quality Filtering: We retain a sample only if all four labeling attempts agree and average confidence exceeds threshold τ :

$$\text{Keep}(q, d) = \left[\bigwedge_{m=1}^4 (l_m = l^*) \right] \cdot \left[\frac{1}{4} \sum_{m=1}^4 p_m(l^*) > \tau \right] \quad (5)$$

where l^* is the consensus label. This strict filtering ensures only high-confidence preference labels enter training, producing final preference dataset $\mathcal{D}_{\text{pref}}$.

C. Multi-Granular Preference Learning and Bidirectional LLM Preference Alignment

In the multi-granular preference learning stage, we capture LLM preferences at three granularities using the preference dataset $\mathcal{D}_{\text{pref}}$. We split $\mathcal{D}_{\text{pref}}$ into two disjoint subsets with a 60:40 ratio: $\mathcal{D}_{\text{pref}}^{(1)}$ for the multi-granular preference learning stage and $\mathcal{D}_{\text{pref}}^{(2)}$ for the bidirectional LLM preference alignment stage, ensuring sufficient data for both training stages while preventing overfitting.

1) Classification Loss (Instance-Level): For each query-document pair (q, d) with label $y \in \{0, 1, 2\}$ (supportive, neutral, or conflicting), we train the reranker as a multi-class classifier:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{|\mathcal{D}_{\text{pref}}^{(1)}|} \sum_{(q,d,y) \in \mathcal{D}_{\text{pref}}^{(1)}} \log p_{\theta}^{(y)}(q, d) \quad (6)$$

where $p_{\theta}^{(y)}(q, d)$ is the predicted probability for class y , obtained via softmax over the three classes.

2) Intra-Query Contrastive Loss (Pair-Level): For documents within the same query, we apply contrastive learning to distinguish supportive documents from others. For each query q , supportive documents serve as positives while neutral and conflicting documents serve as negatives:

$$\mathcal{L}_{\text{intra}} = -\sum_{q \in \mathcal{Q}} \frac{1}{|D_q^+|} \sum_{d^+ \in D_q^+} \log \frac{\exp(h_q \cdot h_{d^+} / \tau_1)}{\sum_{d \in D_q} \exp(h_q \cdot h_d / \tau_1)} \quad (7)$$

where D_q^+ denotes supportive documents (label=0) for query q , h_q and h_d are the embeddings of query and document from the reranker, and τ_1 is the temperature hyperparameter.

3) Inter-Query Contrastive Loss (Set-Level): To capture cross-query patterns, we perform contrastive learning across different query-document pairs in each batch. For each query-document pair (q, d, y) , we pull together pairs with the same label and push apart pairs with different labels:

$$\mathcal{L}_{\text{inter}} = -\sum_{(q,d,y) \in \mathcal{B}} \frac{1}{|\mathcal{B}_y^+|} \sum_{(q',d',y') \in \mathcal{B}_y^+} \log \frac{\exp(h_{qd} \cdot h_{q'd'} / \tau_2)}{\sum_{(q'',d'',y'') \in \mathcal{B}} \exp(h_{qd} \cdot h_{q''d''} / \tau_2)} \quad (8)$$

where $\mathcal{B}$ is the training batch, $\mathcal{B}_y^+$ denotes pairs with the same label y in the batch (excluding (q, d) itself), h_{qd} represents the joint embedding of query-document pair (q, d) , and τ_2 is the temperature hyperparameter.

The total loss combines for this stage these three objectives:

$$\mathcal{L}_{\text{Stage1}} = \lambda_1 \mathcal{L}_{\text{cls}} + \lambda_2 \mathcal{L}_{\text{intra}} + \lambda_3 \mathcal{L}_{\text{inter}} \quad (9)$$

where $\{\lambda_1, \lambda_2, \lambda_3\}$ are hyperparameters balancing the three losses. To avoid potential conflicts and bias in multi-task optimization, we employ Multi-Task Gradient Descent Algorithm with Upper Bound (MGDA-UB) [45] to dynamically compute the optimal weights, ensuring balanced training across all three granularities.

In the bidirectional LLM preference alignment stage, we explicitly model bidirectional preferences using the held-out set $\mathcal{D}_{\text{pref}}^{(2)}$ and the reranker checkpoint from the multi-granular preference learning stage.

Preference Ranking Generation: For each query q with top- k documents D_q , we use LLaMA-2-7B to generate two ranking lists:

- **Positive ranking** π_q^+ : Ranks documents by how well they support correct answer generation
- **Negative ranking** π_q^- : Ranks documents by how much they mislead or contradict

1) Forward Alignment Loss: Promotes documents ranked highly in the positive list using the ListMLE objective:

$$\mathcal{L}_{\text{fwd}} = -\sum_{q \in \mathcal{Q}} \sum_{j=1}^k \log \frac{\exp(s_{\theta}(q, d_{\pi_q^+(j)}))}{\sum_{l=j}^k \exp(s_{\theta}(q, d_{\pi_q^+(l)}))} \quad (10)$$

where $\pi_q^+(j)$ denotes the document at position j in the positive ranking, and $s_{\theta}(q, d)$ is the reranker's score.

2) Backward Alignment Loss: Suppresses documents ranked highly in the negative list:

$$\mathcal{L}_{\text{bwd}} = -\sum_{q \in \mathcal{Q}} \sum_{j=1}^k \log \frac{\exp(-s_{\theta}(q, d_{\pi_q^-(j)}))}{\sum_{l=j}^k \exp(-s_{\theta}(q, d_{\pi_q^-(l)}))} \quad (11)$$

where the negative sign before s_{θ} ensures that documents harmful to generation receive lower scores.

The total loss for this stage is:

$$\mathcal{L}_{\text{Stage2}} = \alpha \mathcal{L}_{\text{fwd}} + \beta \mathcal{L}_{\text{bwd}} \quad (12)$$

where α and β balance the two directional objectives. This bidirectional design addresses the inherent asymmetry in LLMs' positive-negative ranking capabilities. Similarly, we apply MGDA-UB [45] to dynamically optimize the weights between forward and backward alignment losses, achieving effective bidirectional preference modeling.

D. LLM Generation

After obtaining the preference-aligned reranker from the two-stage training, we deploy it in the standard RAG pipeline. For inference, given a query q , we first retrieve top- N documents using a dense retriever, then rerank them using our trained reranker to obtain top- k documents D_{rerank} . These documents are concatenated with the query and fed to the LLM generator to produce the final answer. Our reranked documents substantially improve answer quality by ensuring better alignment with LLM reasoning preferences.

IV. EXPERIMENTS

A. Experimental Setup

Datasets. We conduct experiments on five question answering datasets: CLAPNQ [25], HotpotQA [26], MS_MARCO [27], Natural Questions (NQ) [28], and TriviaQA (TQ) [29]. These datasets cover diverse task types including long-form QA, multi-hop reasoning, passage ranking, open-domain QA, and trivia QA.

Comparison Methods. We compare CQADPA-RAG with representative methods from three categories: (1) *Reranking methods*: RankGPT [46] uses GPT-3.5 to perform list-wise ranking through in-context learning; LRL [47] employs pairwise ranking prompts with large language models; PRP [48] introduces passage-level ranking prompts for finer-grained reranking; RankLLaMA [49] fine-tunes LLaMA models specifically for ranking tasks. These methods represent the reranking-based paradigm that improves retrieval through relevance scoring. (2) *Preference alignment methods*: RAFE [5]

TABLE I
COMPARISON OF LABELING QUALITY BETWEEN MULTI-MODEL ENSEMBLE (MULTI-PRED) AND OUR CONFIDENCE-CALIBRATED APPROACH (MULTI-PRED-CONFIDENT). LABEL 0: SUPPORTIVE, LABEL 1: NEUTRAL, LABEL 2: CONFLICTING.

Dataset	Metric	multi-pred			multi-pred-confident		
		label=0	label=1	label=2	label=0	label=1	label=2
CLAPNQ	Precision	0.9748	0.3793	0.4783	0.9906	0.4545	0.5882
	Recall	0.7838	0.6471	0.6111	0.9292	0.4545	1.0000
	F1	0.8689	0.4783	0.5366	0.9589	0.4545	0.7407
HotpotQA	Precision	0.9512	0.2000	0.8333	0.9568	0.3333	1.0000
	Recall	0.9123	0.5455	0.2778	0.9748	0.3000	0.8182
	F1	0.9313	0.2927	0.4167	0.9657	0.3158	0.9000
MS_MARCO	Precision	0.9583	0.1600	0.4286	0.9843	0.3333	0.7500
	Recall	0.9096	0.2667	0.3750	0.9843	0.3333	0.7500
	F1	0.9333	0.2000	0.4000	0.9843	0.3333	0.7500
NQ	Precision	0.9913	0.1077	0.5000	1.0000	0.2857	0.8333
	Recall	0.6786	0.5385	0.5263	0.9630	0.5000	0.9091
	F1	0.8057	0.1795	0.5128	0.9811	0.3636	0.8696
TQ	Precision	0.9623	0.4390	0.7170	1.0000	0.3333	0.8250
	Recall	0.8718	0.5294	0.7755	0.9583	0.1667	0.9706
	F1	0.9148	0.4800	0.7451	0.9787	0.2222	0.8919

aligns retrievers with generation through instruction-following training; KnowPAT [15] introduces knowledgeable preference alignment for domain-specific QA; DPA-RAG [9] represents the current state-of-the-art with dual preference alignment. These methods explicitly align retrieval with LLM generation needs, providing strong comparison points for our approach. (3) Standard RAG pipeline with DPR retriever serves as the baseline to demonstrate the necessity of reranking or preference alignment. We select these comparison methods because they represent different technical paradigms and the state-of-the-art in their respective categories, enabling comprehensive evaluation of our approach.

We use BGE [49] as the reranker backbone across all experiments which offers strong performance on general embedding tasks and trains efficiently. For retrieval, we employ DPR to obtain top-20 candidate documents from the corpus, then rerank to select top-3 for generation. We chose top-20 as a practical balance between recall and computational efficiency, following common practice in retrieval-augmented generation [9], [14]. Retrieving fewer documents risks missing relevant information while retrieving more increases reranking time without proportional gains. Confidence threshold τ is set to 0.7 based on validation experiments. For query augmentation, GPT-4.1 with temperature 0.7 generates diverse reformulations while maintaining coherence. We use LLaMA-2-7B as the LLM generator for open-source experiments and GPT-4 for closed-source comparisons.

Evaluation Metrics. We use two different metrics to measure how well the system works. Answer Correctness (ACC) checks to see if the answer generated matches the ground truth. This shows how well the RAG system works from retrieval to generation. The Answer Support Rate (ASR) checks to see if the retrieved documents help generate the correct answer by checking to see if important information from the gold answer is present in the top-k retrieved documents. We use GPT-4 to

check if the documents we found have enough information to give the right answer. A higher ASR means that the needs for retrieval and generation are better aligned. This is because documents not only get high ranking scores, but they also really help generate accurate answers. These two metrics give a full picture of how well the generation and retrieval work.

B. Quality of Confidence-Calibrated labeling

We assess the efficacy of our confidence-calibrated labeling mechanism by contrasting two strategies: multi-pred employs a direct multi-model ensemble, whereas multi-pred-confident utilises our confidence calibration and quality filtering methodology. Table I shows the precision, recall, and F1 scores for three label groups across five datasets.

Our confidence-calibrated method gets higher F1 scores for conflicting knowledge (label=2), which is important for getting rid of documents that are wrong. The multi-pred strategy keeps more samples, which helps it remember more, but it also has lower precision, which means that the labels it uses are less accurate and can hurt training later on. Our method keeps high precision while getting competitive recall, which leads to better F1 scores on all datasets.

C. Choosing a Threshold

The effect of the confidence threshold τ on data coverage and labeling quality is shown in Figure 2. While Macro-F1 improves due to higher annotation quality, coverage declines as τ rises because more samples are filtered out. The intrinsic conflict between label reliability and data quantity is reflected in this trade-off: more stringent filtering eliminates low-confidence samples but shrinks the training set. The ideal balance point, which preserves roughly 75% of the data while attaining a Macro-F1 above 0.72, is $\tau = 0.7$. Strong preference signals are provided for model training by this setup, which guarantees an adequate amount of training data with excellent

TABLE II
MAIN EXPERIMENTAL RESULTS ON FIVE QA DATASETS. ACC: ANSWER CORRECTNESS (%). ASR: ANSWER SUPPORT RATE (%).

Method	CLAPNQ		HOTPOTQA		MS_MARCO		NQ		TQ	
	ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR
RankGPT	47.8	85.2	24.5	78.3	27.1	74.5	62.4	88.7	66.3	89.5
LRL	48.1	86.0	25.2	79.8	28.3	76.2	63.4	89.9	67.4	90.8
PRP	51.9	88.5	26.1	81.5	29.0	77.8	66.5	91.2	69.5	92.1
RankLLaMA	52.2	89.0	25.8	80.9	28.6	77.1	67.0	91.8	70.2	92.7
RAFE	50.1	87.3	24.9	79.5	27.8	75.9	64.8	90.3	68.1	91.2
KnowPAT	51.4	88.1	25.6	80.2	28.5	76.8	65.7	90.9	69.3	91.8
DPA-RAG	52.2	91.2	26.8	85.8	29.7	80.5	68.8	94.9	71.8	95.3
CQADPA-RAG	53.5	92.3	27.8	87.2	30.9	82.1	70.2	96.2	73.1	96.5

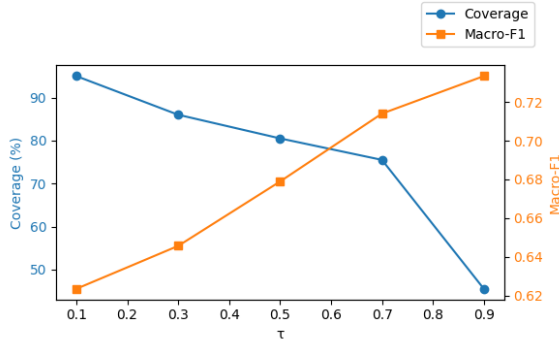


Fig. 2. Impact of confidence threshold τ on data coverage and labeling quality (Macro-F1). We select $\tau = 0.7$ as the optimal balance point.

label quality. We find that thresholds above 0.8 excessively reduce training data, while thresholds below 0.6 retain too many noisy labels, both of which impair model performance.

D. Main Results

The experimental results comparing CQADPA-RAG with state-of-the-art baselines across five datasets are shown in Table II. The efficacy of our strategy is demonstrated that CQADPA-RAG beats all baselines on both ACC and ASR.

CQADPA-RAG achieves improvements of 1.3% ACC on CLAPNQ, 1.0% ACC on HotpotQA, 1.2% ACC on MS_MARCO, 1.4% ACC on NQ, and 1.3% ACC on TQ when compared to the strongest baseline, DPA-RAG. For ACC and ASR, the average improvement across all datasets is 1.24% and 1.66%, respectively. The more noticeable gains in ASR show that our reranked documents are able to find documents that really help answer generation, not just ones that are semantically relevant. This is more in line with LLM reasoning preferences. These results prove that our bidirectional preference alignment and multi-dimensional query augmentation methods work. Our approach's strength and ability to be used in many different situations are shown by the fact that it works better on a wide range of datasets with different characteristics.

TABLE III
ABLATION STUDY RESULTS.

Method	CLAPNQ		HOTPOTQA		MS_MARCO		NQ		TQ	
	ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR	ACC	ASR
Ours	53.5	92.3	27.8	87.2	30.9	82.1	70.2	96.2	73.1	96.5
w/o MPL	49.2	88.1	23.5	82.8	26.7	77.3	65.8	91.9	68.7	92.1
w/o BLPA	51.8	90.5	26.1	85.3	29.2	80.2	68.5	94.6	71.3	95.0

E. Ablation Study

We perform ablation experiments to assess the contribution of each component by individually removing Multi-granular Preference Learning (MPL) and Bidirectional LLM Preference Alignment (BLPA). The results are shown in Table III.

When you take away MPL, the average performance drops by 4.3% ACC and 4.2% ASR. This shows how important it is to capture multi-granular preferences. The average loss of 1.7% ACC and 1.8% ASR when BLPA is removed shows that it is necessary to explicitly model both positive and negative preferences. Both stages help with overall performance, and when they are combined, they work best.

V. CONCLUSION

In this study, we introduce CQADPA-RAG to rectify the discrepancy between retrieval objectives and LLM generation requirements in RAG systems. We use multi-dimensional query augmentation with confidence-calibrated labeling to make high-quality training data. We create four different augmentation strategies that meet a wide range of user information needs. To make sure the quality of the annotations, we use a multi-model ensemble with confidence calibration. We create a two-stage training framework to make sure that retrieval works with LLM preferences. The multi-granular preference learning stage records preferences at the instance, pair, and set levels. The bidirectional LLM preference alignment stage uses dual listwise ranking objectives to model both positive and negative preferences. This fixes the problem of positive-negative ranking asymmetry in current systems. CQADPA-RAG beats the best baselines in both answer accuracy and answer support rate in tests on five question-answering benchmarks. The experimental results show how important it is to match retrieval with LLM-specific preferences when making good RAG systems.

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